

Vo4 Virtual Web Application Team

Description & Eligibility

Teams will create a database-driven website with server-side functionality based on the topic provided. This national event will be submitted and judged virtually. Any middle level, secondary or postsecondary contestant may enter this event. A team will consist of 2-4 contestants. There are no restrictions on the number of entries per chapter or per state. Teams who do not submit an entry following the topic listed below or do not follow the Copyright and/or Fair Use Guidelines will be disqualified.

Topic

MentorshipMatch – Student-to-Professional Mentorship Platform. Your team has been hired by a nonprofit organization to build a web application connecting high school and college students with professionals in their fields of interest. Users can create profiles listing their academic interests and career goals, browse and request mentors, schedule video or chat sessions, and leave reviews after meetings. Required features include user registration and login, mentor/mentee profile management, session booking, and an admin panel to manage users and flag inappropriate activity. Optional enhancements could include a recommendation algorithm, industry-based group forums, resume review requests, and email notifications.

Competition Notes

- AI Policy Level 2 *Permitted with Citation*
- Contest Registration
 - Advisors register contestants for the event using the Membership Registration System, accessible at <https://register.bpa.org>
 - Click on the “Conferences” tab and then select the “2027 National Virtual Events”
 - There will be a \$20 fee for each team entry (*invoiced during event registration*)
 - Event registration can be purchased with your original invoice or with a new invoice after contestant registration
 - The registration deadline is no later than 11:59 p.m. Eastern Time, on December 1, 2026
 - Teams participating in the competition must be registered for the event prior to the submission deadline to be eligible
 - Only the team captain should complete the submission

Technical Pre-Submit Specifications

What to Submit at https://upload.bpa.org	Saved File Name(s)	National Deadline
Submit the following in one combined PDF: 1. Clickable URL to Database Driven Website 2. Written description of the project, source code, and the database schema 3. Works Cited 4. BPA Release Form(s) 5. AI Usage and Documentation Form	Vo4-TeamID.pdf	January 15, 2027, 11:59 p.m. Eastern Time

Contest Competencies

1. Design and implement a database in a structured (SQL) or unstructured (cloud-based, e.g., MongoDB, Firebase) environment to store and manage application data; flat files may not be used for data storage
2. Demonstrate knowledge of server-side scripting and Internet concepts to support dynamic, functional web application behavior
3. Evaluate project requirements and delegate responsibilities effectively within a team, demonstrating the collaboration and communication skills required in a professional business setting

4. Integrate external Application Programming Interfaces (APIs) to extend application functionality and connect to third-party services and data sources
5. Apply back-end database development skills and front-end web design standards to build a complete, functional, and visually consistent full-stack web application
6. Implement structured problem-solving techniques to identify, troubleshoot, and resolve technical challenges throughout the development process
7. Document the process, methodology, and lifecycle of the web application, including an audit trail that demonstrates how the application was planned, built, and iterated

Contest Specifications for this Event

- The team must create a conceptual model for the database identifying relationships between tables in the database. Each team will be expected to articulate the design of the database associated with the web application.
- Each team must create forms and associated considerations needed for the scenario.
- Teams may select the database, web server, and middleware solution they prefer. The source code that powers each team's web application will be scrutinized; therefore, it is imperative that team contestants can answer questions regarding this aspect of the application.
- The event tests the team's ability to write code and incorporate databases. Application frameworks, such as Drupal®, Joomla!®, DotNetNuke® and WordPress® are prohibited. Integrated development environments (IDEs) such as Microsoft® Visual Studio/ASP and Dreamweaver® are acceptable. Packages that assist the developer to create components, menus and similar such as JavaScript®, jQuery® and Ajax are acceptable and encouraged. However, downloading third-party code to create a major component of a web application, such as a shopping cart in an e-commerce application, is *not* permissible. Code must be the original work of the team.
- The website must be available on January 15, 2027. If the ISP is unavailable, the advisor will be contacted and given a reasonable amount of time to solve the problem.
- The site must include name of chapter, team member names, theme, school, city, state and year. This information can be included in any format and location.
- Submit the URL to Database Driven Website including a written description of the application as a PDF file, the source code (as a zipped root folder), the database schema, Works Cited, and signed BPA Release Form(s) in a combined PDF file to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on January 15, 2027.
- Advisors of qualifying teams will be contacted to reserve a time for a videoconference presentation before a panel of judges. The dates and times for presentation reservations will be announced via email notification. Teams will be assigned to sections prior to the preliminary presentation.
- Length of event: no more than three (3) minutes for setup, no more than ten (10) minutes for the presentation, and no more than five (5) minutes for judges' questions.
- Teams with the highest cumulative technical and presentation scores will be invited to attend the National Leadership Conference to be recognized for their outstanding efforts and to participate in the National Showcase.

Common Specifications for all BPA Judged Events

- Team ID is required for all submissions. Materials from non-registered contestants and/or those missing Contestant ID cannot be accepted.
- The team is responsible for securing a BPA Release Form(s) from any person, including themselves, whose image or work is used in the project.
- The team is responsible for citing all sources, including themselves, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Contestants will be disqualified for violations of the guidelines.

- Materials must follow the organization's BPA Graphic Standards and make proper use of the BPA logo and/or organization's name. Refer to the BPA Graphic Standards in the *Style & Reference Manual*.
- Judges' comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

Vo4 Virtual Web Application Team

Technical Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines and Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting (All points or none are awarded by the technical judge)				
Team submitted one (1) combined PDF file containing the URL to the database driven website, written description of the application, source code, and the database schema, Works Cited (formatted according to the <i>Style & Reference Manual</i>), BPA Release Form(s), and AI Usage and Documentation Form.				/10
Page Design	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Layout Clarity	Layout is cluttered or unclear, making navigation and understanding difficult.	Layout is clear and structured, with consistent alignment and effective visual hierarchy.	Layout is highly intuitive and visually engaging, enhancing user understanding through excellent design principles.	/20
Design Aesthetic	Colors and fonts clash or distract, lacking cohesion and visual appeal.	Colors and fonts are appropriately chosen, creating a visually pleasing and consistent design.	Colors and fonts are expertly selected and harmonized, enhancing the overall aesthetic and user experience.	/20
Performance Optimization	Images and files are large or poorly optimized, causing slow load times and reduced performance.	Images and files are reasonably optimized, supporting smooth page performance.	Images and files are expertly optimized, ensuring fast load times and excellent performance across devices.	/20
Writing Quality	Writing is unclear, contains frequent errors, and lacks structure or coherence.	Writing is clear, mostly error-free, and well-organized with appropriate tone and style.	Writing is polished, engaging, and consistently well-structured with excellent clarity and style.	/20
Contrast Between Text and Background	Text blends into the background, making it hard to read.	Text and background have adequate contrast for clear readability.	Contrast is strong and thoughtfully applied, enhancing readability and visual appeal.	/20
Customer Functionality	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Navigational Links	Links are broken or inconsistent.	Links work and follow a consistent style.	Links are fully functional, consistent, and enhance navigation.	/20
Form Instructions and Feedback	No clear instructions or feedback.	Instructions and feedback are present and helpful.	Instructions are clear and feedback is timely and user-friendly.	/20
Cross-Browser/Platform Compatibility	Site fails on some browsers or devices.	Site works reliably across major browsers and devices.	Site performs seamlessly and consistently on all platforms.	/20

Vo4 Virtual Web Application Team

Technical Scoring Rubric (Cont'd)

Search Feature Usability	Search is missing or non-functional.	Search works and returns relevant results.	Search is efficient, accurate, and user-friendly.	/20
Database Development	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Database Use	No database; uses flat files.	Uses a functional database for data storage.	Database is well-integrated and enhances performance.	/20
Conceptual Model Implementation	Model is unclear or incorrectly applied.	Model is correctly implemented and functional.	Model is well-executed and clearly supports the app's structure.	/20
Data Redundancy	Tables contain repeated data.	Redundancy is minimal and controlled	Tables are well-normalized with no unnecessary repetition.	/20
Data Encryption	Sensitive data is unencrypted.	Minimal sensitive data is encrypted.	Encryption is thorough and enhances security.	/20
Database Access Control	No access controls in place.	Access is password-protected and controlled.	Strong access controls ensure secure, role-based access.	/20
Use of Unique Key Concepts	No unique keys used where needed.	Unique keys are correctly applied.	Unique keys are well-structured and optimize data integrity.	/20
Application Design	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
System Diagram Used	No diagram provided, or diagram is unclear, incomplete, or not used in development.	A clear system/application diagram is created and referenced during development.	Diagram is detailed, well-integrated into the development process, and enhances understanding of system architecture.	/20
Session State Management	State is not maintained across sessions or relies solely on cookies without fallback.	State is consistently maintained across sessions using cookies and alternative methods.	State management is robust, secure, and gracefully manages both cookie and non-cookie scenarios.	/20
Sensitive Data Encryption	Sensitive user data is stored or transmitted without encryption, posing security risks.	Highly sensitive user information is encrypted using standard, secure methods.	Encryption is implemented using strong, up-to-date algorithms and integrated seamlessly into the system's security architecture.	/20
Form Validation	Form fields are submitted without validation, leading to potential errors or security issues.	All form fields are validated before submission, ensuring correct and secure input.	Validation is thorough, user-friendly, and includes both client-side and server-side checks.	/20
Use of Queries/Procedures	No use of queries or stored procedures; data access is inefficient or insecure.	Queries and/or stored procedures are used appropriately for data access and manipulation.	Queries and stored procedures are optimized, secure, and demonstrate thoughtful database design.	/20

Vo4 Virtual Web Application Team

Technical Scoring Rubric (Cont'd)

Error Handling	Errors are not captured, or handling is inconsistent and leads to crashes or poor user experience.	Error capture routines are implemented where needed, ensuring stable and predictable behavior.	Error handling is comprehensive, user-friendly, and anticipates edge cases to maintain system reliability.	/20
Administration	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Web-Based Reports	Reports are missing, irrelevant, or do not reflect application data meaningfully.	Appropriate web-based reports are generated, presenting relevant application data clearly.	Reports are well-designed and enhance understanding through dynamic visuals and interactivity.	/20
Admin Data Control	Admin cannot reliably add, modify, or delete data, or interface lacks password protection.	Admin can securely add, modify, and delete data through a password-protected web interface.	Admin interface is secure, intuitive, and supports efficient data management with role-based access or audit features.	/20
Code Documentation	Code lacks comments or documentation, making it difficult to understand or maintain.	Code is adequately documented with clear comments explaining key logic and structure.	Documentation is thorough, consistent, and enhances readability and maintainability across the codebase.	/20
Application Functions	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Report Accuracy	Reports contain incorrect or inconsistent data; results do not reflect actual application behavior.	Reports generate accurate and consistent results based on application data.	Reports are highly accurate, validated against multiple data sources, and reliably reflect real-time or complex data scenarios.	/20
Content Visibility Control	Content visibility (public/private) is not configurable or is inconsistently applied.	Users can designate content as public or private, and the system respects these settings.	Content visibility is well-integrated, secure, and includes intuitive controls or role-based access features.	/20
Admin Functions Operate	Admin functions (add, modify, delete) are incomplete, unreliable, or fail to perform as intended.	Admin can successfully add, modify, and delete data through the interface as described.	Admin functions are fully operational, intuitive, and include safeguards like confirmation prompts or audit logging.	/20
Security Functionality	Security features are missing, improperly implemented, or easily bypassed.	Security functions operate as intended, protecting user data and access appropriately.	Security is robust, proactively mitigates vulnerabilities, and includes advanced features like encryption, role-based access, and audit logging.	/20

Vo4 Virtual Web Application Team

Technical Scoring Rubric (Cont'd)

Core User Features	One or more required features (admin account, user registration, authentication, password protection, content addition) are missing or non-functional.	All required features are implemented and function correctly: admin account, user registration, authentication, password protection, and user content addition.	Required features are fully functional, secure, and enhanced with thoughtful design (e.g., role-based access, password hashing, user-friendly interfaces).	/20
TOTAL TECHNICAL POINTS				/590

Vo4 Virtual Web Application Team

Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines and Artificial Intelligence Usage and Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Evaluation Criteria	Below Expectations 1-5 points	Meets Expectations 6-10 points	Above Expectations 11-20 points	Points Awarded
Development Process	Explanation is unclear, incomplete, or lacks structure; key phases of development are missing.	Clearly explains the design and development process from start to finish with logical flow.	Clearly explains the process, including key decisions, challenges, and solutions.	/20
App via UI	Demonstration is unclear or incomplete; user interface is not effectively used to show application functionality.	Clearly demonstrates how the application works using the user interface; core features are shown effectively.	Demonstration is smooth, engaging, and highlights full functionality through intuitive use of the user interface.	/20
Logical Design	Explanation of the application's logical design and design patterns is unclear, incomplete, or missing.	Provides a clear explanation of the logical design and identifies relevant design patterns used.	Provides a clear, insightful explanation of the logical design and effectively links design patterns to application decisions.	/20
Code Structures	Struggles to articulate how front-end and back-end components work together; explanations are vague or incorrect.	Clearly explains key code structures and their roles in front-end and back-end functionality; demonstrates basic understanding.	Delivers accurate insights into code architecture, clearly connects front-end and back-end logic, and responds confidently to follow-up questions.	/20
Interview Responses	Responses are unclear, off-topic, or lack relevance to the question; struggles to communicate ideas effectively.	Provides clear, relevant, and direct answers; communicates ideas with basic clarity and confidence.	Delivers concise, well-structured, and insightful responses; demonstrates strong communication skills and a deep understanding of the topic.	/20
Writing Accuracy	Frequent errors in grammar, spelling, punctuation, or usage that interfere with clarity and professionalism.	Minor errors may be present but do not hinder understanding; generally clear and correct language use.	Consistently accurate and polished use of grammar, spelling, punctuation, and usage; enhances clarity and professionalism.	/20
Presentation Quality/Style	Presentation lacks structure, clarity, or engagement; delivery is disorganized or difficult to follow.	Presentation is clear, organized, and adequately engaging; communicates ideas effectively.	Presentation is polished, confident, and highly engaging; demonstrates strong delivery, visual clarity, and audience awareness.	/20
Timing	Setup lasted no more than three (3) minutes, and the presentation lasted no more than ten (10) minutes.			/10
Team Members	At least two original members of the team are present during the presentation.			/5
TOTAL PRESENTATION POINTS				/155

Props and/or additional items shall not be used as a basis for scoring.

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES